MLOps Assignment 2

Tracking a Pre-trained Sentiment Model with MLflow

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Objective

To use MLflow to log parameters, metrics and artifacts for a pre-trained Hugging Face model (distilbert-base-uncased-finetuned-sst-2-english), compare multiple runs in the MLflow UI, and report which run performs best. No training is done; only inference is run and tracked. Experiment name: sentiment-inference-lab.

1. Which run had the lowest latency, and why might that be?

run_batch4 had the lowest latency (0.157 s), followed by run_batch8 (0.184 s) and run_batch1 (0.341 s). With batching, the model processes several sentences in one forward pass, so there are fewer inference calls and less per-call overhead (tokenisation, Python and framework overhead). run_batch1 makes 10 separate calls for 10 sentences, so it is the slowest. run_batch4 may have edged out run_batch8 because it ran last, when the model and caches were already warm, and because 10 samples split as 4+4+2 need less padding than one batch of 8 plus a batch of 2. Exact timings vary slightly between executions, but larger batches are generally faster than batch size 1.

2. Did accuracy change between runs? Why or why not?

No. Accuracy stayed at 1.0 (100%) in every run, and average confidence was identical (0.99965). All runs used the same pre-trained model and weights on the same 10 labelled sentences, and inference is deterministic. Batch size only changes how inputs are grouped and how fast they are processed, not the model or its predictions.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

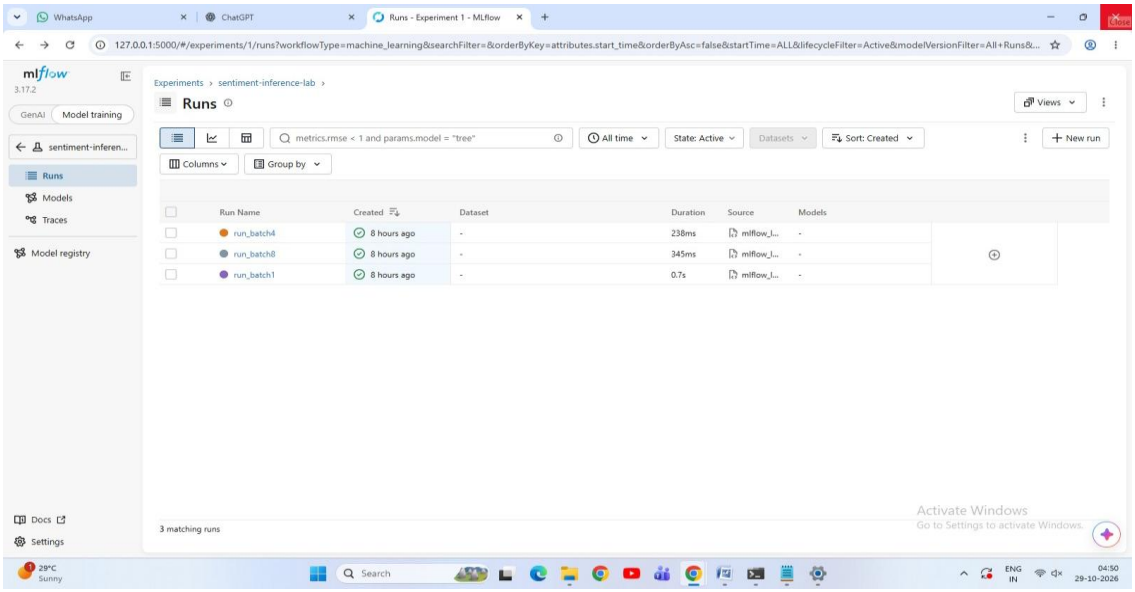
- Reproducibility and record-keeping: MLflow permanently stores the parameters (model_name, batch_size, num_samples), metrics (accuracy, avg_confidence, total_latency_sec) and artifacts (predictions.csv) of every run, whereas terminal output is lost once the window is cleared or closed.
- Easy comparison: runs can be compared side by side in the UI using tables and charts (e.g. parallel coordinates plot), which is much easier than reading printed lines.

Results Summary

Run	batch_size	Accuracy	Avg. confidence	Latency (s)
run_batch1	1	1.0	0.99965	0.341
run_batch8	8	1.0	0.99965	0.184
run_batch4	4	1.0	0.99965	0.157

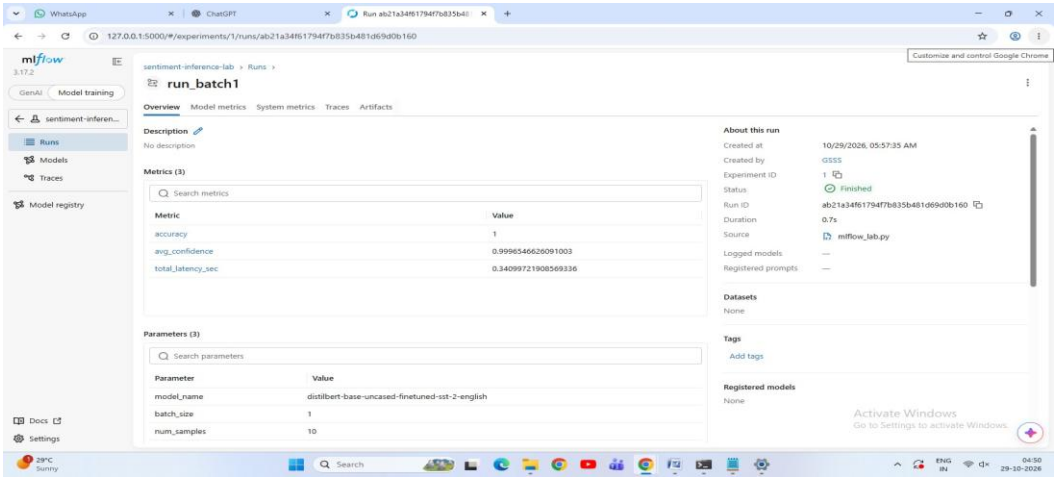
Lowest latency highlighted. Third run added: same model with batch_size=4 (run_batch4).

MLflow Runs Table

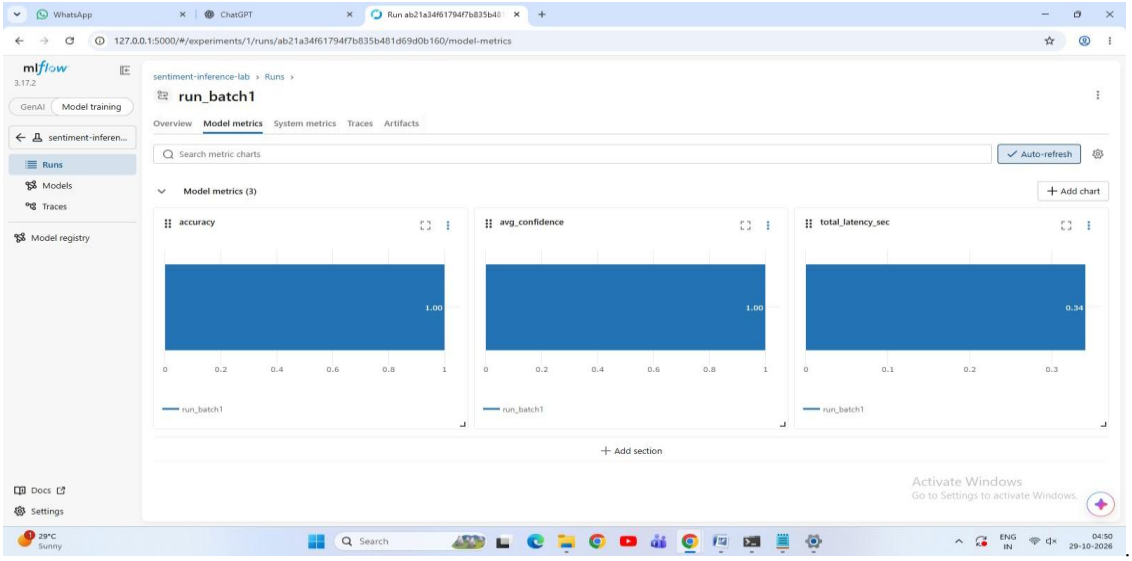


Runs table of experiment sentiment-inference-lab showing run_batch1, run_batch8 and run_batch4 (the third run).

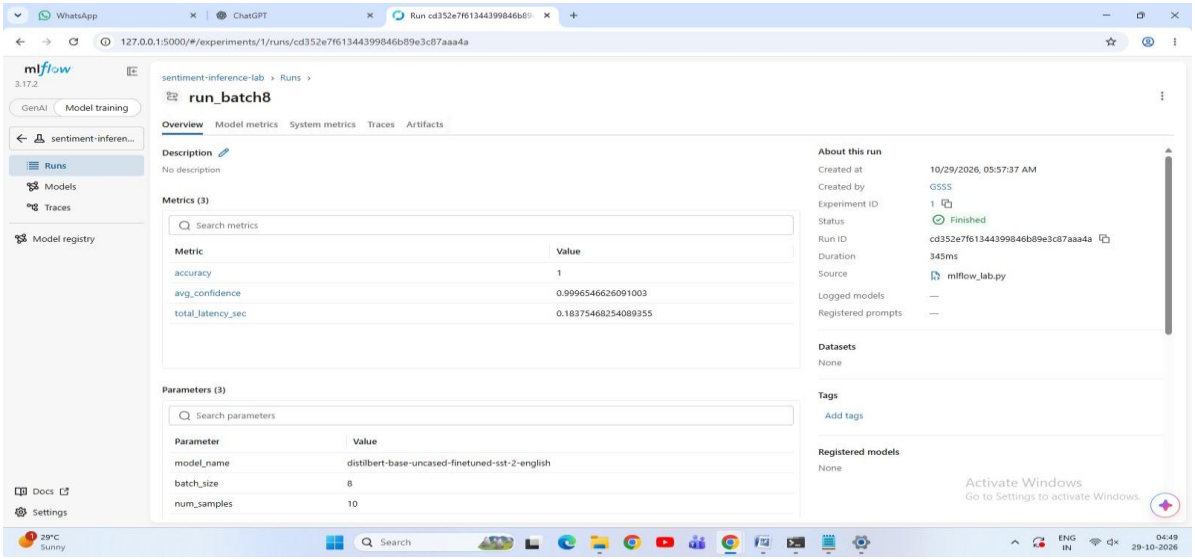
Individual Run Details



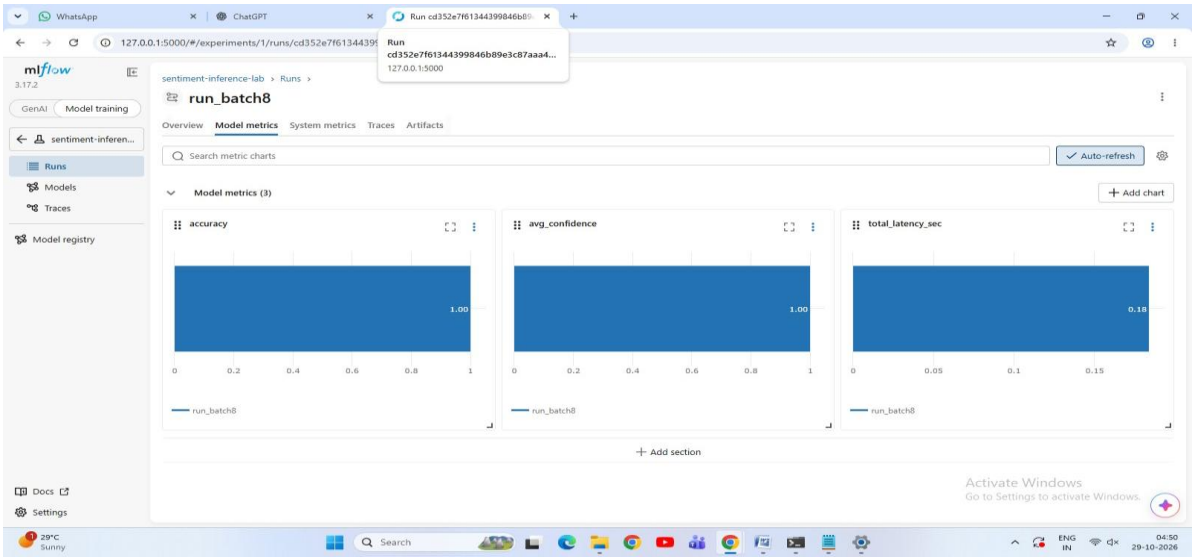
run_batch1: overview with parameters and metrics



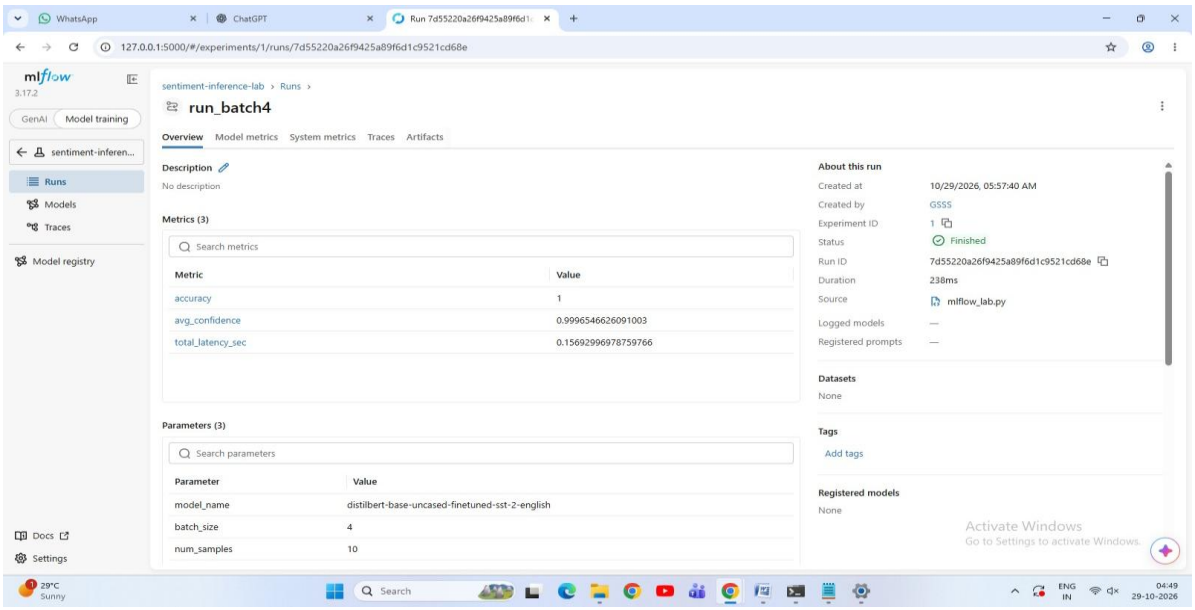
run_batch1: model metrics charts (accuracy, avg_confidence, total_latency_sec).



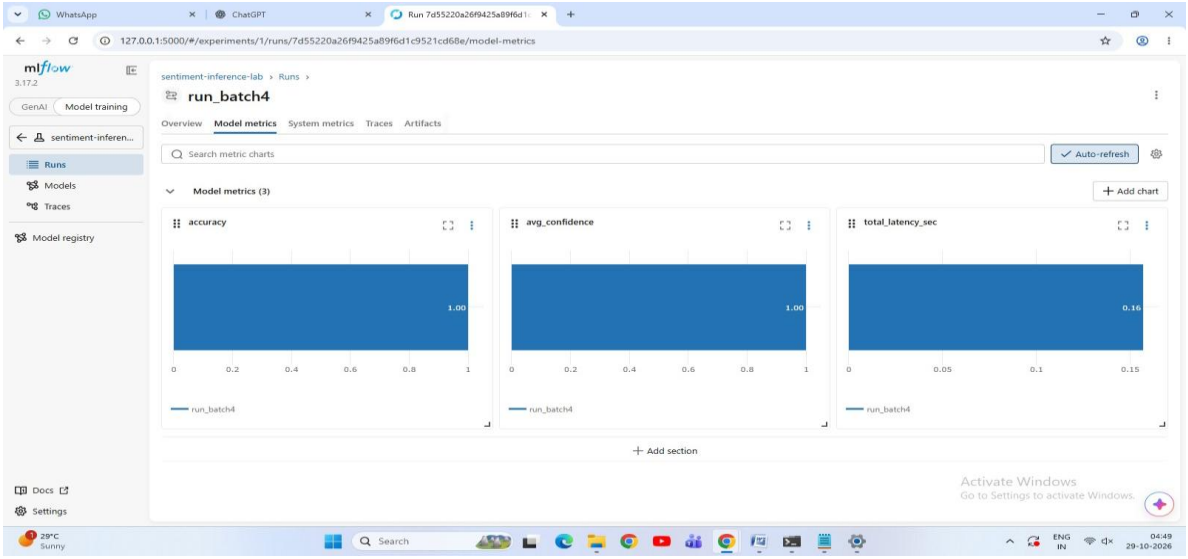
run_batch8: overview with parameters and metrics.



run_batch8: model metrics charts (accuracy, avg_confidence, total_latency_sec).

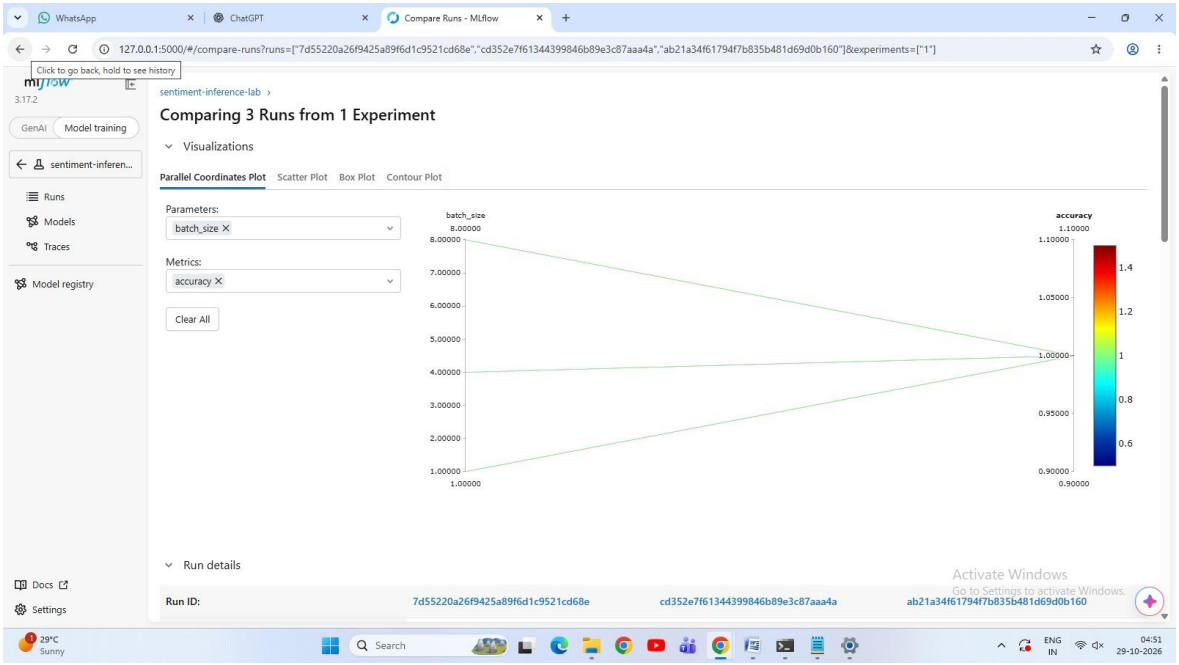


run_batch4: overview with parameters and metrics.



run_batch4: model metrics charts (accuracy, avg_confidence, total_latency_sec).

Compare View



Compare view of all 3 runs (parallel coordinates plot: batch_size vs accuracy).